Composing transformations

1

a Reflection and translation

b Rotation and translation

2

a Reflection and translation

b Rotation and translation

3 Shape A

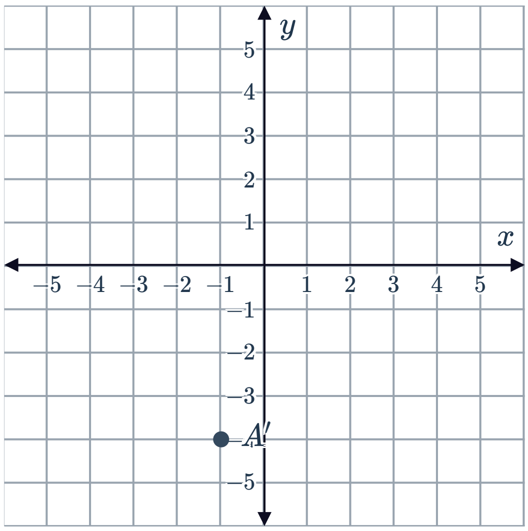
4 Triangle W

5 Square A

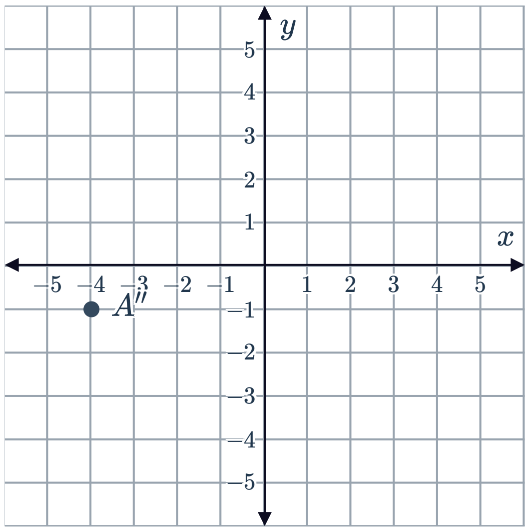
6 $CBAD$

7

a

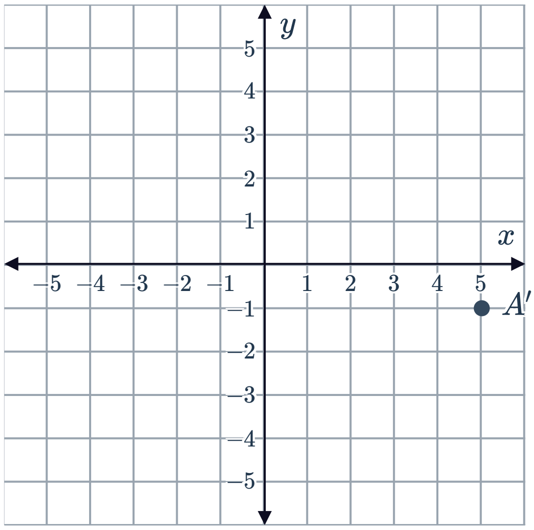


b

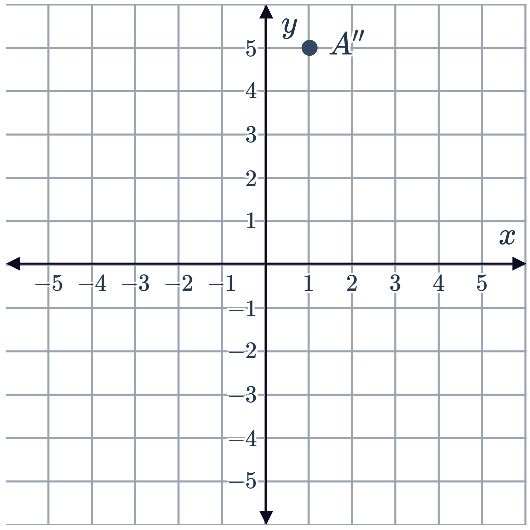


8

a

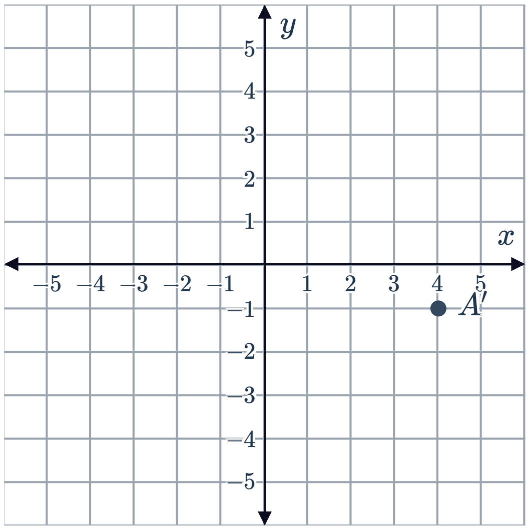


b

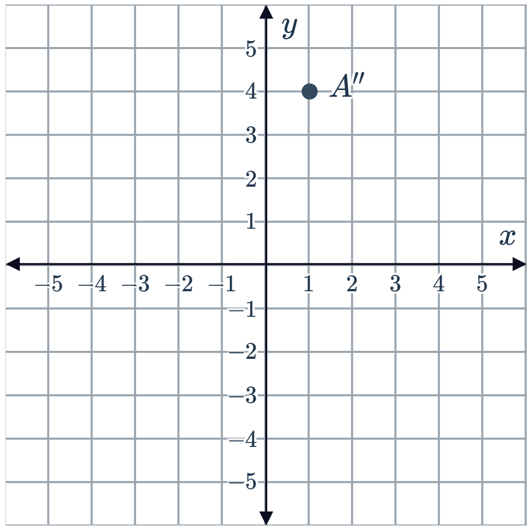


9

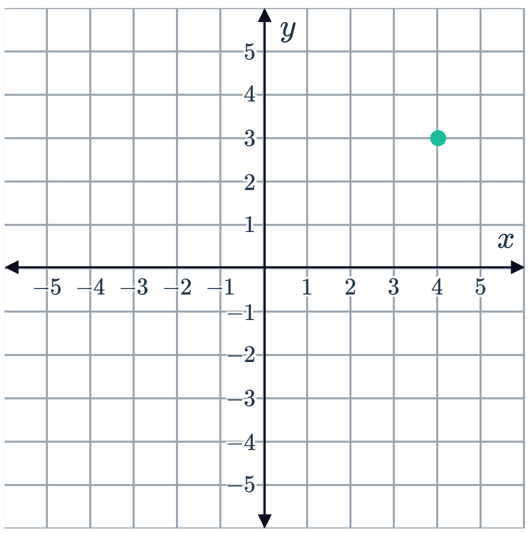
a



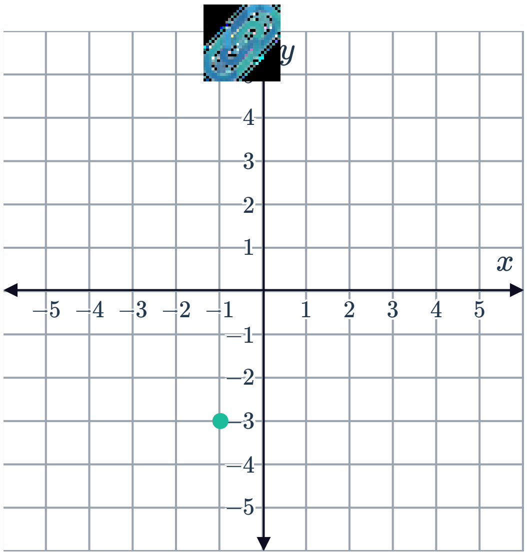
b



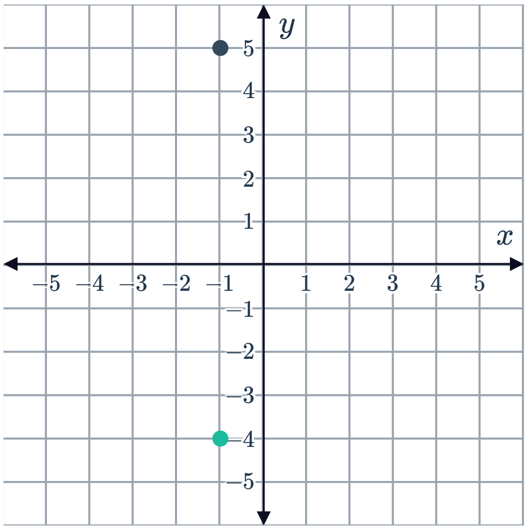
10



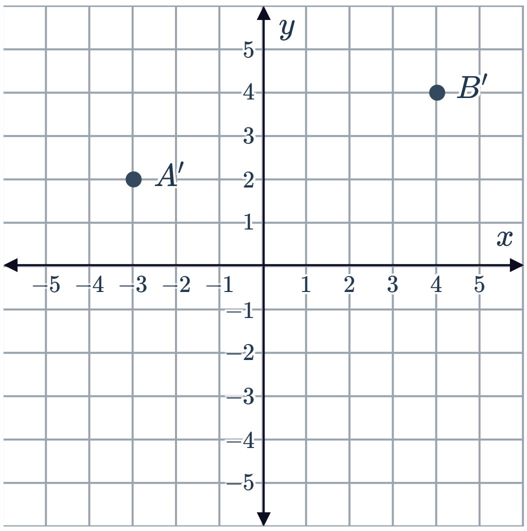
11



12

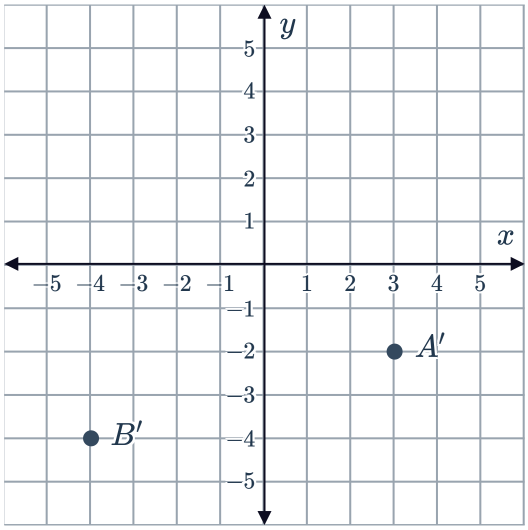


13



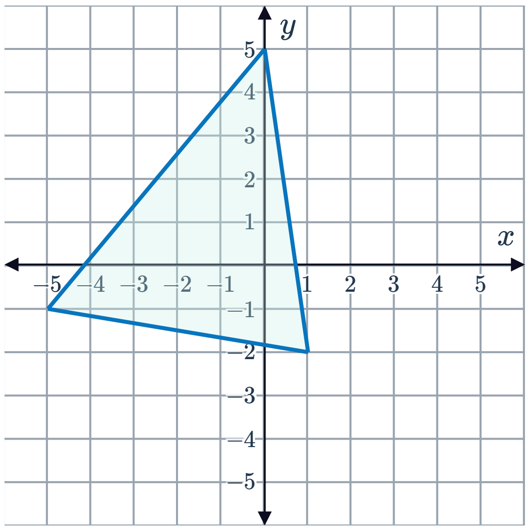
14

a

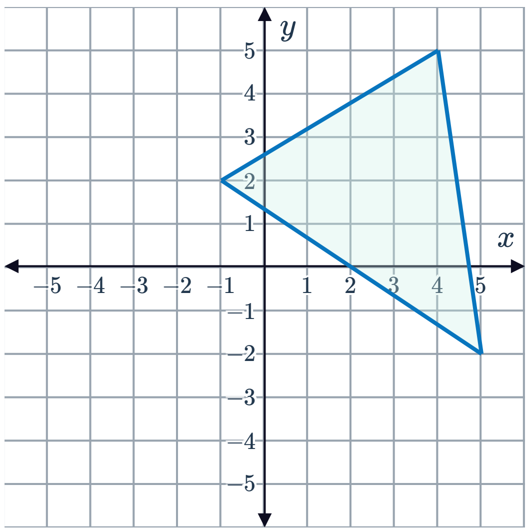


b A reflection across the x-axis.

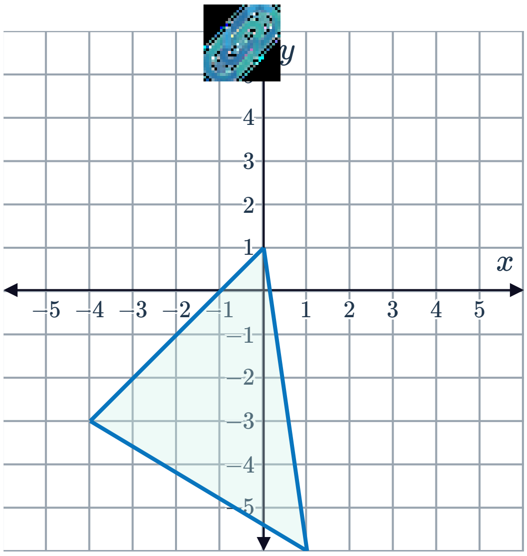
15



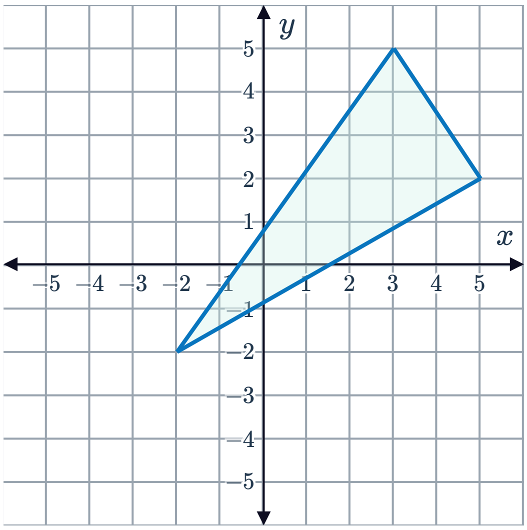
16



17

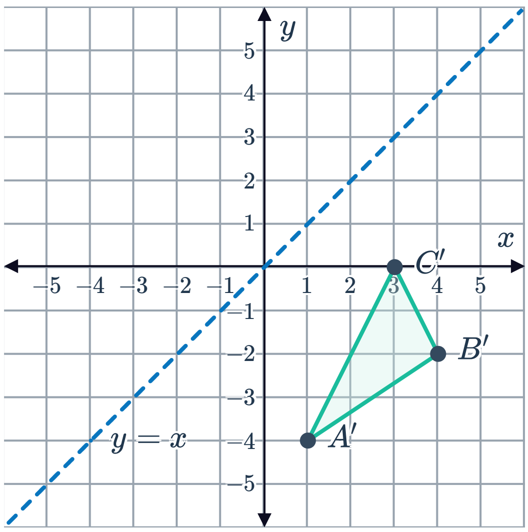


18



19

a



b A 90° rotation anticlockwise about the origin.

20

a $B\,(4,-1)$

b $y=1$

21 A reflection across the y-axis, and a transl[illegible]10 units left and 11 units up.

22 A translation 11 units right and 3 units up, and a reflection across the x-axis.

23 A reflection across the x-axis, and a translation 10 units left and 8 units down.

24 A translation 9 units right and 8 units up, and a reflection across the x-axis.

25
- **a** $B\left(6,10\right)$
- **b** 16 units

26 $\left(9,10\right)$

27 $\left(1,2\right)$

28
- **a** $\left(-5,6\right)$
- **b** $\left(4,-7\right)$

29 $\left(1,3\right)$

30 $\left(11,7\right)$

31
- **a** $A'\left(-6,-1\right)$
- **b** A reflection across the y-axis.

32
- **a** $A'\left(4,-4\right),B'\left(-8,4\right)$
- **b** A reflection across the y-axis.

33
- **a** $A'\left(-1,-2\right),B'\left(-7,3\right)$
- **b** A reflection across the x-axis.

34 $A'\left(-8,2\right),B'\left(-5,9\right),C'\left(-14,6\right)$

35 $A'\left(5,4\right),B'\left(-1,-1\right),C'\left(2,-10\right)$

36 $A'\left(2,-8\right),B'\left(-4,5\right),C'\left(-3,6\right)$

37 $A'\left(9,-5\right),B'\left(-10,1\right),C'\left(-10,-3\right)$

39 $A'(-1,-9), B'(-7,-2), C'(-6,4)$

40 $(-x,-y)$

41 Reflecting the point across both the x and y-axes.

42 $(a,-b)$

43

- **a** $B(0,9)$
- **b** 9 unit translation horizontally to the right, and 9 unit translation vertically upwards.

44

- **a** $C(5,-6)$
- **b** Rotation 180° about the origin.